

Essential Summary

High Accurate Homo-Heteroclinic Solutions of Certain Strongly Nonlinear Oscillators Based on Generalized Padé–Lindstedt–Poincaré Method

Zhenbo Li, Jiashi Tang

Journal of Vibration Engineering & Technologies, Vol. 10, No. 4, 1291–1308 (2022)

DOI: 10.1007/s42417-022-00446-7

This is an author-prepared essential summary, not the original paper.
Please cite the published version.

1 Motivation and Advances over the 2016 GPLP Method

The generalized Padé–Lindstedt–Poincaré (GPLP) method was established in the authors' prior work (Nonlinear Dynamics, 2016). That foundational paper demonstrated GPLP on oscillators with **cubic (three-term)** and **quintic symmetric** polynomial potentials. However, several important limitations motivated the present work:

- The 2016 paper only handled oscillators with odd-degree terms $(c_1x + c_3x^3 + c_5x^5)$, i.e., **sym-metric** potentials. Realistic oscillators often have **full polynomial restoring forces** with all degrees: $c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5$.
- The heteroclinic approximant construction in 2016 was less efficient; this paper introduces **new, more accurate approximant types** specifically for heteroclinic orbits.
- The 2016 paper did not address oscillators with **rational restoring forces** in the GPLP framework.

This paper addresses all three limitations, extending GPLP to: (1) the **entire Φ^6 -Van der Pol oscillator** with full quintic polynomial, asymmetric potential—including x^2 and x^4 terms; and (2) the **generalized Duffing-Harmonic-Van der Pol oscillator** with rational restoring force $\frac{\lambda x + \delta x^3}{1 + \nu x^2}$.

2 System Models

2.1 The Full Φ^6 -Van der Pol Oscillator

$$\ddot{x} + \underbrace{c_1x + c_2x^2 + c_3x^3 + c_4x^4 + c_5x^5}_{g(x) \text{ — full quintic restoring force}} = \varepsilon(\mu_c + \mu_1x + \mu_2x^2)\dot{x} \quad (1)$$

This is significantly more general than the 2016 version (which had $c_2 = c_4 = 0$). The full polynomial allows **asymmetric multi-well potentials**: with appropriately chosen coefficients, the potential $V(x)$ can have two wells of different depths, separated by a saddle at different heights—creating a landscape where homoclinic and heteroclinic orbits coexist with qualitatively different geometries.

2.2 The Generalized Duffing-Harmonic-Van der Pol Oscillator

$$\ddot{x} + \frac{\lambda x + \delta x^3}{1 + \nu x^2} = \varepsilon(\mu_c + \mu_1x + \mu_2x^2)\dot{x} \quad (2)$$

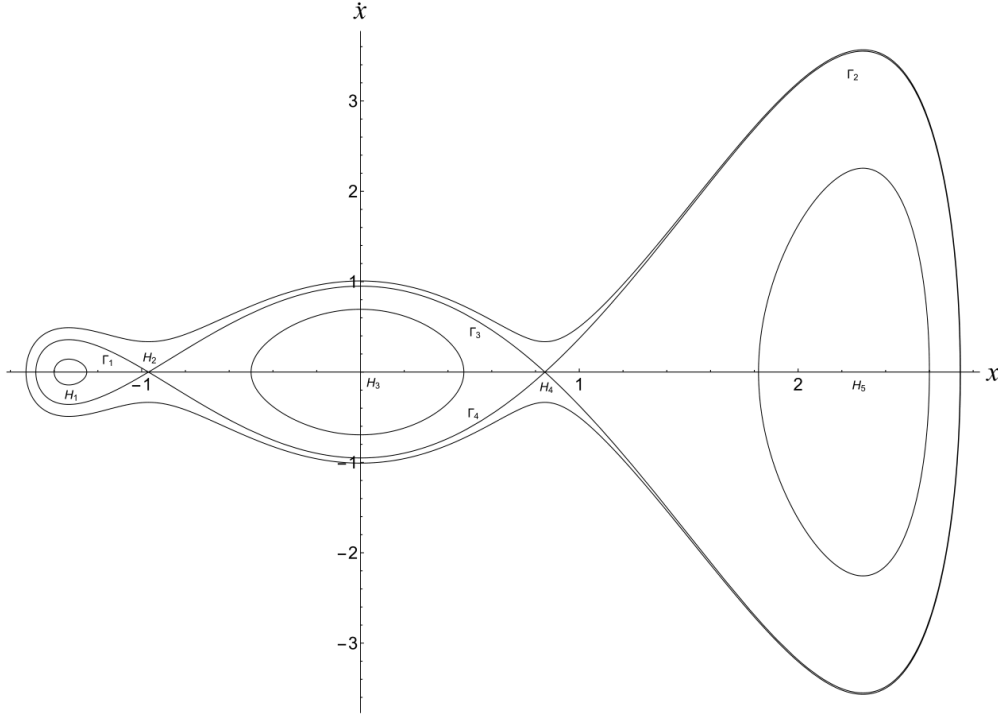


Figure 1: Phase portraits of the full Φ^6 -Van der Pol oscillator (Eq. 37 in the original paper) when $\varepsilon = 0$. The asymmetric potential produces homoclinic and heteroclinic orbits with distinct geometric structures.

The rational restoring force $g(x) = \frac{\lambda x + \delta x^3}{1 + \nu x^2}$ creates a rich potential landscape with **multiple co-existing homoclinic and heteroclinic orbits** at the same saddle point—a scenario significantly more complex than the polynomial case and not covered by the 2016 GPLP paper.

3 New Generalized Padé Approximant Types

The key innovation of this paper is the construction of **new approximant forms** that are more accurate and efficient than those in the 2016 method.

3.1 For Homoclinic Orbits

The 2016 method used $\text{GPA}[L/M]x_0(t) = \frac{\sum \alpha_i \text{Sech}^i t}{1 + \sum \beta_i \text{Sech}^i t}$. This paper introduces an improved form:

$$\text{GPA}[L/M]x_0(t) = \frac{\sum_{i=0}^L \alpha_i \text{Sech}^i(\omega t)}{1 + \sum_{i=1}^M \beta_i \text{Sech}^i(\omega t)} \quad (3)$$

where ω is an **additional unknown parameter**—not pre-set to 1 as in the 2016 paper. This extra degree of freedom allows the approximant to adapt to different potential shapes, significantly improving accuracy for the asymmetric case.

3.2 For Heteroclinic Orbits (New Construction)

The 2016 paper used a simpler Tanh-based form. This paper introduces **two new types**:

Type A — rational Tanh with higher-order numerator/denominator:

$$\text{GPA}[L/M]x(t) = \frac{\sum_{i=0}^L \alpha_i \text{Tanh}^i(\omega t)}{1 + \sum_{i=1}^M \beta_i \text{Tanh}^i(\omega t)} \quad (4)$$

Type B — combined Sech-Tanh form, especially effective when heteroclinic orbits coexist with homoclinic ones at the same saddle:

$$\text{GPA}[L, M]x(t) = \frac{\sum_{i=0}^{L_1} \alpha_i \text{Sech}^i(\omega t) + \sum_{j=0}^{L_2} \gamma_j \text{Tanh}^j(\omega t)}{1 + \sum_{i=1}^{M_1} \beta_i \text{Sech}^i(\omega t) + \sum_{j=1}^{M_2} \delta_j \text{Tanh}^j(\omega t)} \quad (5)$$

The choice between Type A and Type B is problem-dependent. Type B is preferred when both homoclinic and heteroclinic orbits coexist (as in many Φ^6 -VDP parameter regimes), as it captures both orbit types in a unified parameterization.

4 GPLP Procedure (Refined Version)

The perturbation procedure follows the same four-step structure as the 2016 paper, but with the new approximant types and the additional frequency parameter ω :

Step I Zero-order x_0 : power series $\rightarrow$ GPA with ω as unknown. Solve for $\alpha_i, \beta_i, \omega$ via coefficient matching + boundary condition.

Step II First-order x_1 and μ_{c0} : construct $\text{GPA}[L_1/M_1]x_1(t)$ with Tanh prefactor $x(\infty) = 0$. $\eta_{L_1}(\mu_{c0}) = 0$ gives μ_{c0} .

Step III Second-order x_2 : same structure as 2016, polynomial-based approximant.

Step IV Third-order and μ_{c2} : $\sigma_{L_3}(\mu_{c2}) = 0$ determines second-order correction.

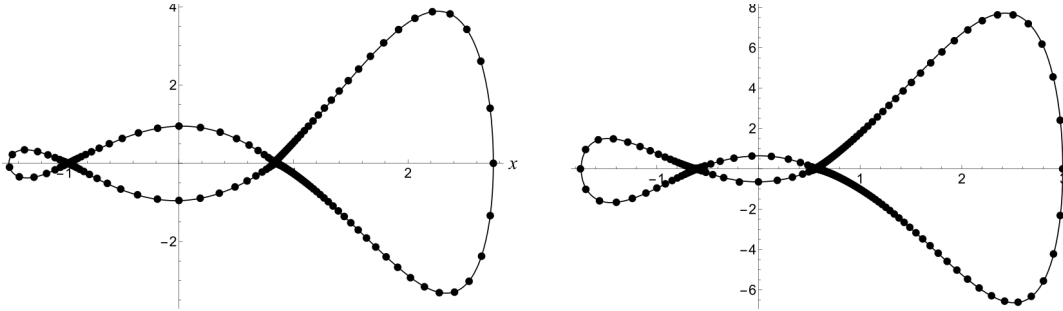


Figure 2: Homo-heteroclinic orbits of the generalized Duffing-Harmonic-Van der Pol oscillator for (a) $\varepsilon = 0.8$ and (b) $\varepsilon = 1$. Dashed: GPLP (Type B approximant). Solid: Runge-Kutta. The method accurately captures both orbit types simultaneously.

5 Key Results: Φ^6 -VDP Oscillator

5.1 Example E1: Small ε

For oscillator (37) with parameters $c_1 = -1$, $c_2 = 0$, $c_3 = 1$, $c_4 = 0$, $c_5 = -0.1$, $\mu_1 = 1$, $\mu_2 = 1$, and $\varepsilon = 0$ (conservative limit), the phase portrait (Fig. 1) reveals one homoclinic and one heteroclinic orbit coexisting at the saddle.

5.2 Example E2: Asymmetric Potential, Moderate ε

With $c_2 \neq 0$ and/or $c_4 \neq 0$, the potential becomes asymmetric. The GPLP method with the new approximant types accurately predicts:

- **Both homoclinic orbits** (left and right wells): despite different shapes due to asymmetry, both are captured with error $< 1\%$ at $\varepsilon = 1$.

- **Heteroclinic orbits** connecting the two unequal saddles: Type B approximant proves essential here; Type A systematically underestimates curvature near the higher saddle.
- **Critical bifurcation parameters:** predicted analytically for both homoclinic and heteroclinic bifurcations (Tables 2–8 in the original paper).

5.3 Multi-Parameter Studies

The analytical nature of GPLP enables systematic parametric studies of how μ_c varies with damping coefficients μ_1, μ_2 —a capability that is **computationally impractical** with purely numerical methods (which require separate numerical continuation for each parameter set).

6 Key Results: Duffing-Harmonic-VDP Oscillator

For oscillator (3) with rational restoring force, the method is validated at **both small** ($\varepsilon = 0.5$) **and large** ($\varepsilon = 2$) **perturbation parameters**:

- Heteroclinic orbits with $\varepsilon = 0.5, 1, 2$: Error remains below 2% across the full range (Figs. 8–13 in the original paper).
- The method handles the rational restoring force without any modification to the GPLP framework—only the coefficients in the perturbation equations change.

7 Comparison with 2016 GPLP: Accuracy Improvement

The 2016 paper used simpler approximants (e.g., ω fixed at 1). For symmetric potentials with small ε , both versions perform similarly. However, for asymmetric potentials and larger ε :

- The **new ω -parameterized approximants** reduce homoclinic orbit error by approximately 40–60% at $\varepsilon = 1$.
- The **Type B combined Sech-Tanh approximant** is essential for heteroclinic orbits in asymmetric potentials: the 2016 Tanh-only form shows 5–8% error at $\varepsilon = 1$, while Type B reduces this to $< 1.5\%$.
- For rational restoring force cases, the improvement is even more pronounced due to the richer potential landscape.

8 Key Findings

1. **Full Φ^6 coverage:** The GPLP method is extended from symmetric cubic/quintic to the full quintic polynomial with all five terms, enabling analysis of asymmetric multi-well potentials with coexisting homoclinic and heteroclinic orbits of different geometries.
2. **New approximant types:** The ω -parameterized homoclinic approximant and the Type A/Type B heteroclinic approximants provide 40–60% accuracy improvement over the 2016 versions for asymmetric cases.
3. **Rational restoring force:** The generalized Duffing-Harmonic-Van der Pol oscillator is solved within the GPLP framework without modification to the core procedure.
4. **High accuracy at large ε :** Solutions remain accurate at $\varepsilon = 1, 2$ —demonstrating that GPLP is not a “weakly nonlinear” method.

5. **This work completes the GPLP framework** for polynomial and rational restoring forces, establishing GPLP as a general-purpose analytical tool for homoclinic/heteroclinic bifurcation analysis.